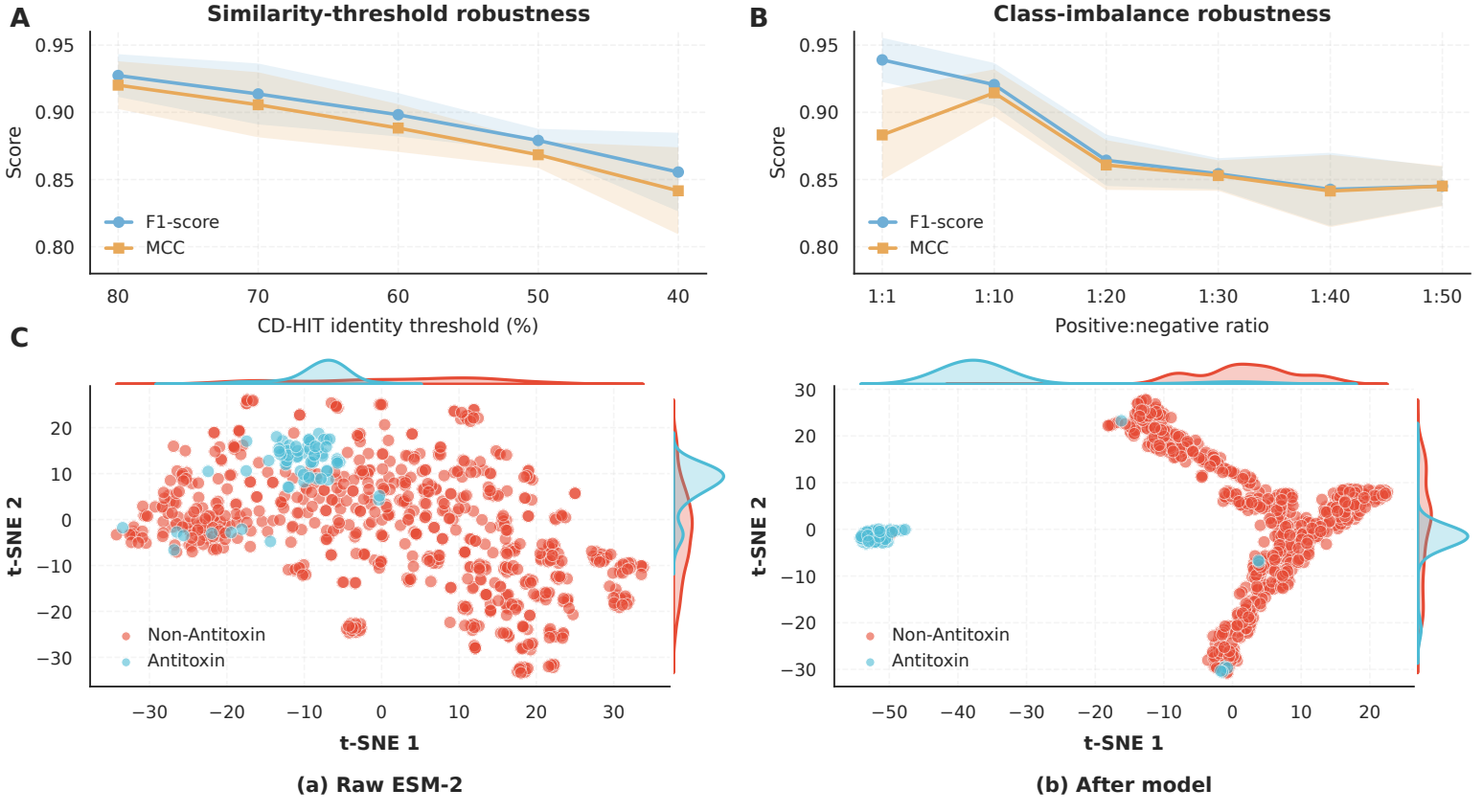

A
Similarity-threshold robustness
Score
0.95
0.90
0.85
0.80
F1-score
MCC
80
70
60
50
40
CD-HIT identity threshold (%)
B
Class-imbalance robustness
Score
0.95
0.90
0.85
0.80
F1-score
MCC
1:1
1:10
1:20
1:30
1:40
1:50
Positive:negative ratio
C
t-SNE 2
20
10
0
−10
−20
−30
Non-Antitoxin
Antitoxin
−30
−20
−10
0
10
20
30
t-SNE 1
(a) Raw ESM-2
t-SNE 2
30
20
10
0
−10
−20
−30
Non-Antitoxin
Antitoxin
−50
−40
−30
−20
−10
0
10
20
t-SNE 1
(b) After model